Energy Institute AVIFF - Worked Examples : EI AVIFF Screening Report

Line listing

Record ID	P&ID	Line reference	Description	Notes	Pipe details	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material	Stream	Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]	% of stream	Flow Velocity [m/s]	Energy ρv^2 [kg/m s²]	Max. Span	Stiffness category	LOF : Stiff	LOF : Medium-Stiff	LOF : Medium	LOF : Flexible	Flow-induced turbulence	Mechanical excitation	Pulsation Recip/PD
1	Figure D-1	Stream 4	Upstream of E401		14" STD Carbon Steel	14"	STD	355.6	9.525		0.00	Carbon Steel	Stream 4(gas)	Stream 4	gas	0.97	59358	141.9	18	26	.00002	425.7984016073446	1.22	23.22	100	10.0	1909	11	medium-stiff	0.01	0.01	0.03	0.11	0.01		
2	Figure D-1	Stream 5	Downstream of E401, upstream of 402		14" STD Carbon Steel	14"	STD	355.6	9.525		0.00	Carbon Steel	Stream 5(gas)	Stream 5	gas	0.97	59358	30	26	24.5	.00001	375.64219086193657	1.3	23.22	100	7.0	1321	11	medium-stiff	0.00	0.01	0.01	0.05	0.01		
3	Figure D-1	Stream 6	Downstream of V402, upstream of K402		14" STD Carbon Steel	14"	STD	355.6	9.525		0.00	Carbon Steel	Stream 6(gas)	Stream 6	gas	0.97	53482	30	23	24.5	.00001	394.05474816543466	1.34	21.75	100	7.0	1213	11	medium-stiff	0.00	0.01	0.01	0.05	0.01	0.40	
4	Figure D-1	Stream 7	Downstream of K402		8" 120 Carbon Steel	8"	120	219.075	18.263		0.00	Carbon Steel	Stream 7(gas)	Stream 7	gas	0.97	53482	136.7	62	88	.00002	456.4	1.33	21.75	100	9.0	5197	8	medium-stiff	0.01	0.02	0.04	0.12	0.02	0.40	
5	Figure D-1	Recycle line (compressor discharge)	From downstream of K402 back to upstream of recycle valve		8" 120 Carbon Steel	8"	120	219.075	18.263		0.00	Carbon Steel	Stream 7(gas)	Stream 7	gas	0.97	53482	136.7	62	88	.00002	456.4	1.33	21.75	100	9.0	5197	8	medium-stiff	0.01	0.02	0.04	0.12	0.02	0.40	
6	Figure D-1	Recycle line (compressor suction)	From upstream of recycle valve to upstream of E401		6" STD Carbon Steel	6"	STD	168.275	7.112		0.00	Carbon Steel	Recycle low pressure(gas)	Recycle low pressure	gas	0.97	53482	141.9	18	26	.00002	425.7984016073446	1.22	23.22	100	44.0	35294	7	medium-stiff	0.14	0.28	0.54	1.90	0.28		
7	Figure D-1	Relief line (upstream of PSV)	Downstream of K402 to upstream of PSV		4" 120 Carbon Steel	4"	120	114.3	11.125		0.00	Carbon Steel	Relief high pressure(gas)	Relief high pressure	gas	0.97	53482	136.7	62	88	.00002	456.4	1.33	21.75	100	36.0	80379	6	medium-stiff	0.18	0.34	0.60	2.04	0.34	0.40	
8	Figure D-1	Relief line (downstream of PSV)	Downstream of PSV to flare		6" STD Carbon Steel	6"	STD	168.275	7.112		0.00	Carbon Steel	Relief low pressure(gas)	Relief low pressure	gas	0.97	53482	88	4	1	.00001	428.4	1.33	21.75	100	199.0	158823	7	medium-stiff	0.43	0.88	1.71	6.06	0.88		
9	Figure D-3	Stream 1	From production header		20" STD Carbon Steel	20"	STD	508	9.525		0.00	Carbon Steel	Stream 1 (revised)(multiphase)	Stream 1 (revised)	multiphase	0.8	855976	50	136	6.5	0	310.80779873363696	1.01	28.09	100	9.0	11791	9	stiff	0.26	0.63	1.19	5.47	0.26		
10	Figure D-3	Stream 2	Gas to LP compressor		12" STD Carbon Steel	12"	STD	323.85	9.525		0.00	Carbon Steel	Stream 2 (revised)(gas)	Stream 2 (revised)	gas	0.97	27180	50	5	6.5	.00001	389.22911440640155	1.24	21.99	100	21.0	2141	12	medium	0.01	0.01	0.02	0.09			
11	Figure D-3	Stream 3	Oil to cooler		10" STD Carbon Steel	10"	STD	273.05	9.271		0.00	Carbon Steel	Stream 3 (revised)(liquid)	Stream 3 (revised)	liquid	0.02	303180	50	930	6.5	.3935	275.28103012583387	1.08	38.29	100	2.0	2947	11	medium	0.07	0.15	0.27	1.12			
12	Figure D-3	Stream 4	Produced water		10" STD Carbon Steel	10"	STD	273.05	9.271		0.00	Carbon Steel	Stream 4 (revised)(liquid)	Stream 4 (revised)	liquid	0.02	525616	50	1000	6.5	.00055	1293	1.16	18.05	100	3.0	8237	19	flexible	0.20	0.42	0.75	3.12	3.12		

Stream listing

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Stream 4	gas	0.97	59358	141.9	18	26	.00002	425.7984016073446	1.22	23.22
Stream 5	gas	0.97	59358	30	26	24.5	.00001	375.64219086193657	1.3	23.22
Stream 6	gas	0.97	53482	30	23	24.5	.00001	394.05474816543466	1.34	21.75

Stream 7	gas	0.97	53482	136.7	62	88	.00002	456.4	1.33	21.75
Recycle low pressure	gas	0.97	53482	141.9	18	26	.00002	425.7984016073446	1.22	23.22
Relief high pressure	gas	0.97	53482	136.7	62	88	.00002	456.4	1.33	21.75
Relief low pressure	gas	0.97	53482	88	4	1	.00001	428.4	1.33	21.75
Stream 1 (original)	multiphase	0.8	461764	50	930	6.5	0	310.80779873363696	1.01	28.09
Stream 2 (original)	gas	0.97	27180	50	5	6.5	.00001	389.22911440640155	1.24	21.99
Stream 3 (original)	liquid	0.02	303180	50	930	6.5	.3935	275.28103012583387	1.08	38.29
Stream 4 (original)	liquid	0.02	131404	50	988	6.5	.00055	415.52530955512975	1.16	18.05
Stream 1 (revised)	multiphase	0.8	855976	50	136	6.5	0	310.80779873363696	1.01	28.09
Stream 2 (revised)	gas	0.97	27180	50	5	6.5	.00001	389.22911440640155	1.24	21.99
Stream 3 (revised)	liquid	0.02	303180	50	930	6.5	.3935	275.28103012583387	1.08	38.29
Stream 4 (revised)	liquid	0.02	525616	50	1000	6.5	.00055	1293	1.16	18.05

Detailed listing

Line: Stream 4

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
1	Figure D-1	Stream 4	Upstream of E401	100	14"	STD	355.6	9.525		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Stream 4	gas	0.97	59358	141.9	18	26	.00002	425.7984016073446	1.22	23.22

(FIT) Flow-induced turbulence (EI AVIFF Guidelines T2.2)	0.01
Maximum pipe span [m](Using standard EI 2008 Steel curves)	10.84
Natural Frequency [Hz]	8.29
Calculation Method	exact

Flow-induced pulsation from deadlegs (EI AVIFF Guidelines T2.2) 0.00

Branch ref	Branch nominal bore	Branch schedule	Branch internal diameter [mm]	Throat diameter [mm]	Throat edge radius [mm]	Deadleg length [m]	Type	Upstream straight length [m]	EI LOF	Advanced LOF (Non Structurally Resonant)	Advanced LOF (Structurally Resonant)

Line: Stream 5

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
2	Figure D-1	Stream 5	Downstream of E401, upstream of 402	100	14''	STD	355.6	9.525		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Stream 5	gas	0.97	59358	30	26	24.5	.00001	375.64219086193657	1.3	23.22

(FIT) Flow-induced turbulence (EI AVIFF Guidelines T2.2)	0.01
Maximum pipe span [m](Using standard EI 2008 Steel curves)	10.84
Natural Frequency [Hz]	8.25
Calculation Method	exact

Flow-induced pulsation from deadlegs (EI AVIFF Guidelines T2.2) 0.00

Branch ref	Branch nominal bore	Branch schedule	Branch internal diameter [mm]	Throat diameter [mm]	Throat edge radius [mm]	Deadleg length [m]	Type	Upstream straight length [m]	EI LOF	Advanced LOF (Non Structurally Resonant)	Advanced LOF (Structurally Resonant)

Line: Stream 6

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
3	Figure D-1	Stream 6	Downstream of V402, upstream of K402	100	14"	STD	355.6	9.525		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Stream 6	gas	0.97	53482	30	23	24.5	.00001	394.05474816543466	1.34	21.75

(FIT) Flow-induced turbulence (EI AVIFF Guidelines T2.2)	0.01
Maximum pipe span [m](Using standard EI 2008 Steel curves)	10.84
Natural Frequency [Hz]	8.26
Calculation Method	exact

Mechanical Excitation (EI AVIFF Guidelines T2.3) 0.40

Equipment tag	Type	Size	Analysis Passed	Support	LOF
	large motor/alternator				0.40
	centrifugal compressor				0.20

Flow-induced pulsation from deadlegs (EI AVIFF Guidelines T2.2) 0.00

Branch ref	Branch nominal bore	Branch schedule	Branch internal diameter [mm]	Throat diameter [mm]	Throat edge radius [mm]	Deadleg length [m]	Type	Upstream straight length [m]	EI LOF	Advanced LOF (Non Structurally Resonant)	Advanced LOF (Structurally Resonant)

Thermowells (ASME PTC 19.3 TW - 2016) 0.29

Item tag	Type	Connection method	Root diameter (A) [mm]	Tip diameter (B) [mm]	Fillet radius at base (b) [mm]	Fillet radius at step (b_s) [mm]	Internal bore diameter (d) [mm]	Unsupported length (L) [mm]	Reduced diameter length (L_s) [mm]	Minimum tip thickness (t) [mm]	Shielded length (L_0) [mm]	Damping ratio [Note 1]	Velocity override (V) [m/s] [Note 2]	Material override [Note 3]	Natural frequency override [Note 4]	LOF
TT-001	Tapered	Flanged/Lap-joint	26.5	18	1	0	8	225	0	5	1					0.29

Line: Stream 6 Thermowell TT-001 (ASME PTC 19.3-TW-2016)

Thermowell Details	
Type	Tapered
Connection	Flanged/Lap-joint
Root diameter (A)	26.5 mm
Tip diameter (B)	18 mm
Fillet radius at base (b)	1 mm
Internal bore diameter (d)	8 mm
Unsupported length (L)	225 mm
Minimum tip thickness (t)	5 mm
Shielded length (L_0)	1 mm
Material	Carbon Steel
Process Data	
Stream	Stream 6
Phase	gas
Operating Density	23 kg/m³
Operating Pressure	24.5 Bara
Operating Temperature	30 °C
Fluid dynamic viscosity	.00001 Pa.s
Fluid velocity	7.26 m/s
Pipe information	
Nominal Bore	14''
Schedule	STD
Outside diameter	355.6 mm
Wall thickness	10 mm
Internal Cladding	0 mm
Material	Carbon Steel

Frequency	
Reynolds Number	300600
Strouhal Number	0.20
Scruton Number	1.334
In Situ Resonant Frequency	424.6 Hz
Vortex shedding Frequency	80.6 Hz
Frequency Ratio	0.19
Critial Velocity	17.42 m/s
Material Properties	
Max Allowable Working Stress	138 MPa
Modulus of elasticity	202 GPa
Reference modulus of elasticity	203 GPa
Fatigue Stress Limit	48.3 MPa
Material Density	7750 kg/m³
Sensor Average Density	2700 kg/m³
Cyclic Stresses at inline resonance	
Cyclic drag stress $S_{O,max}$	200.7 MPa
Cyclic stress limit	48.2 MPa
Steady state stresses at design velocity	
Calculated Von Mises Stress	0.5 MPa
Max. Allowable Von Mises Stress	206.9 MPa
Dynamic stresses at design velocity	
Combined Drag and Lift Stress	0.4 MPa
Dynamic stress limit	48.2 MPa
Pressure	
Max Allowable Stem Pressure	47.2 MPa
Max Allowable Tip Pressure	414.4 MPa

Assessment Result	PASS

Line: Stream 7

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
4	Figure D-1	Stream 7	Downstream of K402	100	8"	120	219.075	18.263		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Stream 7	gas	0.97	53482	136.7	62	88	.00002	456.4	1.33	21.75

(FIT) Flow-induced turbulence (EI AVIFF Guidelines T2.2)	0.02
Maximum pipe span [m](Using standard EI 2008 Steel curves)	8.32
Natural Frequency [Hz]	8.20
Calculation Method	exact

Mechanical Excitation (EI AVIFF Guidelines T2.3) 0.40

Equipment tag	Type	Size	Analysis Passed	Support	LOF
	large motor/alternator				0.40
	centrifugal compressor				0.20

Flow-induced pulsation from deadlegs (EI AVIFF Guidelines T2.2) 1.00

Branch ref	Branch nominal bore	Branch schedule	Branch internal diameter [mm]	Throat diameter [mm]	Throat edge radius [mm]	Deadleg length [m]	Type	Upstream straight length [m]	EI LOF	Advanced LOF (Non Structurally Resonant)	Advanced LOF (Structurally Resonant)
Recycle line	8"	120	182.55		20.2	5.1			1.00	NaN	NaN

(SBC) Small bore connections (EI AVIFF Guidelines) 0.66

SBC Ref.	SBC tag	Fitting type	Length	Valves	SBC diameter	SBC location	Mainline LOF	Mainline schedule	LOF Geom/loc/mod	LOF	Walkdown comments	Survey Link
Example D5	Pressure tapping	weldolet	< 600mm	1 (Less than #900)	1" DN25	mid span	1.00	80	0.66/1.00/0.66	0.66		

Line: Recycle line (compressor discharge)

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
5	Figure D-1	Recycle line (compressor discharge)	From downstream of K402 back to upstream of recycle valve	100	8"	120	219.075	18.263		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Stream 7	gas	0.97	53482	136.7	62	88	.00002	456.4	1.33	21.75

(FIT) Flow-induced turbulence (EI AVIFF Guidelines T2.2)	0.02
Maximum pipe span [m](Using standard EI 2008 Steel curves)	8.32
Natural Frequency [Hz]	8.20
Calculation Method	exact

Mechanical Excitation (EI AVIFF Guidelines T2.3) 0.40

Equipment tag	Type	Size	Analysis Passed	Support	LOF
	large motor/alternator				0.40
	centrifugal compressor				0.20

Flow-induced pulsation from deadlegs (EI AVIFF Guidelines T2.2) 0.00

Branch ref	Branch nominal bore	Branch schedule	Branch internal diameter [mm]	Throat diameter [mm]	Throat edge radius [mm]	Deadleg length [m]	Type	Upstream straight length [m]	EI LOF	Advanced LOF (Non Structurally Resonant)	Advanced LOF (Structurally Resonant)

(AIV) High frequency acoustic excitation (EI AVIFF Guidelines T2.7) 0.29

Valve tag	Rated mass flow rate [kg/h]	Pressure upstream [bara]	Pressure downstream [bara]	Temperature upstream [°C]	Molecular weight [g/mol]	Low Noise trim [dB]	PWL [dB]	LOF	Notes
Recycle	53482	88	26	136.7	21.75	30	129.36	0.29	30dB low noise trim

Line: Recycle line (compressor suction)

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
6	Figure D-1	Recycle line (compressor suction)	From upstream of recycle valve to upstream of E401	100	6"	STD	168.275	7.112		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Recycle low pressure	gas	0.97	53482	141.9	18	26	.00002	425.7984016073446	1.22	23.22

(FIT) Flow-induced turbulence (EI AVIFF Guidelines T2.2)	0.28
Maximum pipe span [m](Using standard EI 2008 Steel curves)	7.27
Natural Frequency [Hz]	8.62
Calculation Method	exact

Flow-induced pulsation from deadlegs (EI AVIFF Guidelines T2.2) 0.00

Branch ref	Branch nominal bore	Branch schedule	Branch internal diameter [mm]	Throat diameter [mm]	Throat edge radius [mm]	Deadleg length [m]	Type	Upstream straight length [m]	EI LOF	Advanced LOF (Non Structurally Resonant)	Advanced LOF (Structurally Resonant)

(AIV) High frequency acoustic excitation (EI AVIFF Guidelines T2.7) 0.29

Valve tag	Rated mass flow rate [kg/h]	Pressure upstream [bara]	Pressure downstream [bara]	Temperature upstream [°C]	Molecular weight [g/mol]	Low Noise trim [dB]	PWL [dB]	LOF	Notes
Recycle	53482	88	26	136.7	23.22	30	129.02	0.29	30dB low noise trim

Line: Relief line (upstream of PSV)

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
7	Figure D-1	Relief line (upstream of PSV)	Downstream of K402 to upstream of PSV	100	4"	120	114.3	11.125		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m^3]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Relief high pressure	gas	0.97	53482	136.7	62	88	.00002	456.4	1.33	21.75

(FIT) Flow-induced turbulence (EI AVIFF Guidelines T2.2)	0.34
Maximum pipe span [m](Using standard EI 2008 Steel curves)	6.09
Natural Frequency [Hz]	7.89
Calculation Method	exact

Mechanical Excitation (EI AVIFF Guidelines T2.3) 0.40

Equipment tag	Type	Size	Analysis Passed	Support	LOF
	large motor/alternator				0.40
	centrifugal compressor				0.20

Flow-induced pulsation from deadlegs (EI AVIFF Guidelines T2.2) 1.00

Branch ref	Branch nominal bore	Branch schedule	Branch internal diameter [mm]	Throat diameter [mm]	Throat edge radius [mm]	Deadleg length [m]	Type	Upstream straight length [m]	EI LOF	Advanced LOF (Non Structurally Resonant)	Advanced LOF (Structurally Resonant)
PSV bypass	2"	160	42.85		9.7	.3			1.00	NaN	NaN

Valve transients (EI AVIFF Guidelines T2.8) 3.77

Item tag	Item	Transient Mechanism	Upstream pressure [bara]	Downstream pressure [bara]	Upstream temperature [°C]	Is Valve downstream Of Pump?	Pump head at zero flow [bara]	Valve closure time [s]	Upstream pipe length [m]	Stiffness category	Valve Opening Fmax [kN]	Valve Closing Fmax [kN]	Valve Closing Pmax [MPa]	Valve opening LOF	Valve closing LOF	Max LOF
	Unknown				136.7						6.28	NaN	NaN	3.77	0.00	3.77

Line: Relief line (downstream of PSV)

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
8	Figure D-1	Relief line (downstream of PSV)	Downstream of PSV to flare	100	6"	STD	168.275	7.112		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Relief low pressure	gas	0.97	53482	88	4	1	.00001	428.4	1.33	21.75

(FIT) Flow-induced turbulence (EI AVIFF Guidelines T2.2)	0.88
Maximum pipe span [m](Using standard EI 2008 Steel curves)	7.27
Natural Frequency [Hz]	8.66
Calculation Method	medium-stiff

Flow-induced pulsation from deadlegs (EI AVIFF Guidelines T2.2) 1.00

Branch ref	Branch nominal bore	Branch schedule	Branch internal diameter [mm]	Throat diameter [mm]	Throat edge radius [mm]	Deadleg length [m]	Type	Upstream straight length [m]	EI LOF	Advanced LOF (Non Structurally Resonant)	Advanced LOF (Structurally Resonant)
PSV bypass	2"	80	49.25		10.2	1.1			1.00	5.37	268.35

(AIV) High frequency acoustic excitation (EI AVIFF Guidelines T2.7) 1.48

Valve tag	Rated mass flow rate [kg/h]	Pressure upstream [bara]	Pressure downstream [bara]	Temperature upstream [°C]	Molecular weight [g/mol]	Low Noise trim [dB]	PWL [dB]	LOF	Notes
Relief	53482	99	1	137	21.75	0	164.69	1.48	

Valve transients (EI AVIFF Guidelines T2.8) 1.77

Item tag	Item	Transient Mechanism	Upstream pressure [bara]	Downstream pressure [bara]	Upstream temperature [°C]	Is Valve downstream Of Pump?	Pump head at zero flow [bara]	Valve closure time [s]	Upstream pipe length [m]	Stiffness category	Valve Opening Fmax [kN]	Valve Closing Fmax [kN]	Valve Closing Pmax [MPa]	Valve opening LOF	Valve closing LOF	Max LOF
	Not a valve				137.7						6.29	NaN	NaN	1.77	0.00	1.77

Line: Stream 1

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
9	Figure D-3	Stream 1	From production header	100	20"	STD	508	9.525		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Stream 1 (revised)	multiphase	0.8	855976	50	136	6.5	0	310.80779873363696	1.01	28.09

(FIT) Flow-induced turbulence (EI AVIFF Guidelines T2.2)	0.26
Maximum pipe span [m](Using standard EI 2008 Steel curves)	9.03
Natural Frequency [Hz]	15.72
Calculation Method	exact

Line: Stream 2

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
10	Figure D-3	Stream 2	Gas to LP compressor	100	12"	STD	323.85	9.525		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Stream 2 (revised)	gas	0.97	27180	50	5	6.5	.00001	389.22911440640155	1.24	21.99

Line: Stream 3

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
11	Figure D-3	Stream 3	Oil to cooler	100	10"	STD	273.05	9.271		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Stream 3 (revised)	liquid	0.02	303180	50	930	6.5	.3935	275.28103012583387	1.08	38.29

Line: Stream 4

Record Id	P&ID	Line reference	Description	% of stream	Nominal bore	Schedule	OD [mm]	WT [mm]	Cladding [mm]	Pipe pressure rating [bara]	Material
12	Figure D-3	Stream 4	Produced water	100	10"	STD	273.05	9.271		0.00	Carbon Steel

Description	Phase	Gas Void Fraction	Mass flow rate [kg/h]	Temperature [°C]	Density [kg/m³]	Static pressure [bara]	Dynamic viscosity [Pa.s]	Speed of sound [m/s]	Gamma (c_p/c_v) [#]	Molecular weight [g/mol]
Stream 4 (revised)	liquid	0.02	525616	50	1000	6.5	.00055	1293	1.16	18.05

(FIT) Flow-induced turbulence (EI AVIFF Guidelines T2.2)	3.12
Maximum pipe span [m](Using standard EI 2008 Steel curves)	18.61
Natural Frequency [Hz]	1.59
Calculation Method	exact

Valve transients (EI AVIFF Guidelines T2.8) 0.06

Item tag	Item	Transient Mechanism	Upstream pressure [bara]	Downstream pressure [bara]	Upstream temperature [°C]	Is Valve downstream Of Pump?	Pump head at zero flow [bara]	Valve closure time [s]	Upstream pipe length [m]	Stiffness category	Valve Opening Fmax [kN]	Valve Closing Fmax [kN]	Valve Closing Pmax [MPa]	Valve opening LOF	Valve closing LOF	Max LOF
FCV1002	Globe	Both	6.5	6.5		No		5	2		0.00	0.23	0.00	0.00	0.06	0.06

(FIT) Flow-induced turbulence (EI AVIFF Guidelines T2.2)

Maximum pipe span [m](Using standard EI 2008 Steel curves)	10.84
Natural Frequency [Hz]	8.29
Calculation Method	exact
Maximum pipe span [m](Using standard EI 2008 Steel curves)	10.84
Natural Frequency [Hz]	8.25
Calculation Method	exact
Maximum pipe span [m](Using standard EI 2008 Steel curves)	10.84
Natural Frequency [Hz]	8.26
Calculation Method	exact
Maximum pipe span [m](Using standard EI 2008 Steel curves)	8.32
Natural Frequency [Hz]	8.20
Calculation Method	exact
Maximum pipe span [m](Using standard EI 2008 Steel curves)	8.32
Natural Frequency [Hz]	8.20
Calculation Method	exact
Maximum pipe span [m](Using standard EI 2008 Steel curves)	7.27
Natural Frequency [Hz]	8.62
Calculation Method	exact
Maximum pipe span [m](Using standard EI 2008 Steel curves)	6.09
Natural Frequency [Hz]	7.89
Calculation Method	exact
Maximum pipe span [m](Using standard EI 2008 Steel curves)	7.27
Natural Frequency [Hz]	8.66
Calculation Method	medium-stiff
Maximum pipe span [m](Using standard EI 2008 Steel curves)	9.03
Natural Frequency [Hz]	15.72
Calculation Method	exact
Maximum pipe span [m](Using standard EI 2008 Steel curves)	18.61
Natural Frequency [Hz]	1.59
Calculation Method	exact

Mechanical Excitation (EI AVIFF Guidelines T2.3)

Record Id	P&ID	Line reference	Description	Equipment tag	Type	Size	Analysis Passed	Support	LOF
3	Figure D-1	Stream 6	Downstream of V402, upstream of K402		large motor/alternator				0.40
3	Figure D-1	Stream 6	Downstream of V402, upstream of K402		centrifugal compressor				0.20
4	Figure D-1	Stream 7	Downstream of K402		large motor/alternator				0.40
4	Figure D-1	Stream 7	Downstream of K402		centrifugal compressor				0.20
5	Figure D-1	Recycle line (compressor discharge)	From downstream of K402 back to upstream of recycle valve		large motor/alternator				0.40
5	Figure D-1	Recycle line (compressor discharge)	From downstream of K402 back to upstream of recycle valve		centrifugal compressor				0.20
7	Figure D-1	Relief line (upstream of PSV)	Downstream of K402 to upstream of PSV		large motor/alternator				0.40
7	Figure D-1	Relief line (upstream of PSV)	Downstream of K402 to upstream of PSV		centrifugal compressor				0.20

Pulsation from reciprocating items (EI AVIFF Guidelines T2.4)

Pulsation from rotating stall (EI AVIFF Guidelines T2.5)

Flow-induced pulsation from deadlegs (EI AVIFF Guidelines T2.2)

Record Id	P&ID	Line reference	Description	Branch ref	Branch nominal bore	Branch schedule	Branch internal diameter [mm]	Throat diameter [mm]	Throat edge radius [mm]	Deadleg length [m]	Type	Upstream straight length [m]	EI LOF	Advanced LOF (Non Structurally Resonant)	Advanced LOF (Structurally Resonant)
4	Figure D-1	Stream 7	Downstream of K402	Recycle line	8"	120	182.55		20.2	5.1			1.00	NaN	NaN
7	Figure D-1	Relief line (upstream of PSV)	Downstream of K402 to upstream of PSV	PSV bypass	2"	160	42.85		9.7	.3			1.00	NaN	NaN
8	Figure D-1	Relief line (downstream of PSV)	Downstream of PSV to flare	PSV bypass	2"	80	49.25		10.2	1.1			1.00	5.37	268.35

Pulsation from corrugated pipes (EI Subsea AVIFF Guidelines E4.3)

Slug flow (EI AVIFF Guidelines - Specialist)

(AIV) High frequency acoustic excitation (EI AVIFF Guidelines T2.7)

Record Id	P&ID	Line reference	Description	Valve tag	Rated mass flow rate [kg/h]	Pressure upstream [bara]	Pressure downstream [bara]	Temperature upstream [°C]	Molecular weight [g/mol]	Low Noise trim [dB]	PWL [dB]	LOF	Notes
5	Figure D-1	Recycle line (compressor discharge)	From downstream of K402 back to upstream of recycle valve	Recycle	53482	88	26	136.7	21.75	30	129.36	0.29	30dB low noise trim
6	Figure D-1	Recycle line (compressor suction)	From upstream of recycle valve to upstream of E401	Recycle	53482	88	26	136.7	23.22	30	129.02	0.29	30dB low noise trim
8	Figure D-1	Relief line (downstream of PSV)	Downstream of PSV to flare	Relief	53482	99	1	137	21.75	0	164.69	1.48	

Valve transients (EI AVIFF Guidelines T2.8)

Record Id	P&ID	Line reference	Description	Item tag	Item	Transient Mechanism	Upstream pressure [bara]	Downstream pressure [bara]	Upstream temperature [°C]	Is Valve downstream Of Pump?	Pump head at zero flow [bara]	Valve closure time [s]	Upstream pipe length [m]	Stiffness category	Valve Opening Fmax [kN]	Valve Closing Fmax [kN]	Valve Closing Pmax [MPa]	Valve opening LOF	Valve closing LOF	Max LOF
7	Figure D-1	Relief line (upstream of PSV)	Downstream of K402 to upstream of PSV		Unknown				136.7						6.28	NaN	NaN	3.77	0.00	3.77
8	Figure D-1	Relief line (downstream of PSV)	Downstream of PSV to flare		Not a valve				137.7						6.29	NaN	NaN	1.77	0.00	1.77
12	Figure D-3	Stream 4	Produced water	FCV1002	Globe	Both	6.5	6.5		No		5	2		0.00	0.23	0.00	0.00	0.06	0.06

Cavitation and flashing (EI AVIFF Guidelines T2.9)

Thermowells (ASME PTC 19.3 TW - 2016)

Record Id	P&ID	Line reference	Description	Item tag	Type	Connection method	Root diameter (A) [mm]	Tip diameter (B) [mm]	Fillet radius at base (b) [mm]	Fillet radius at step (b_s) [mm]	Internal bore diameter (d) [mm]	Unsupported length (L) [mm]	Reduced diameter length (L_s) [mm]	Minimum tip thickness (t) [mm]	Shielded length (L_0) [mm]	Damping ratio [Note 1]	Velocity override (V) [m/s] [Note 2]	Material override [Note 3]	Natural frequency override [Note 4]	LOF
3	Figure D-1	Stream 6	Downstream of V402, upstream of K402	TT-001	Tapered	Flanged/Lap-joint	26.5	18	1	0	8	225	0	5	1					0.29

Thermowells (ASME PTC 19.3 TW - 2016) Summary Listing

Record Id	P&ID	Line reference	Description	Item tag	Type	Connection method	In Situ Resonant Frequency	Vortex shedding Frequency	Frequency Ratio	Critial Velocity	Cyclic drag stress $S_{O,max}$	Cyclic stress limit	LOF
3	Figure D-1	Stream 6	Downstream of V402, upstream of K402	TT-001	Tapered	Flanged/Lap-joint	424.6	80.6	0.190	17.4	200.66	48.16	0.29

Line: Stream 6 Thermowell TT-001 (ASME PTC 19.3-TW-2016)

Thermowell Details	
Type	Tapered
Connection	Flanged/Lap-joint
Root diameter (A)	26.5 mm
Tip diameter (B)	18 mm
Fillet radius at base (b)	1 mm
Internal bore diameter (d)	8 mm
Unsupported length (L)	225 mm
Minimum tip thickness (t)	5 mm
Shielded length (L_0)	1 mm
Material	Carbon Steel
Process Data	
Stream	Stream 6
Phase	gas
Operating Density	23 kg/m³
Operating Pressure	24.5 Bara
Operating Temperature	30 °C
Fluid dynamic viscosity	.00001 Pa.s
Fluid velocity	7.26 m/s
Pipe information	
Nominal Bore	14''
Schedule	STD
Outside diameter	355.6 mm
Wall thickness	10 mm
Internal Cladding	0 mm
Material	Carbon Steel

Frequency	
Reynolds Number	300600
Strouhal Number	0.20
Scruton Number	1.334
In Situ Resonant Frequency	424.6 Hz
Vortex shedding Frequency	80.6 Hz
Frequency Ratio	0.19
Critial Velocity	17.42 m/s
Material Properties	
Max Allowable Working Stress	138 MPa
Modulus of elasticity	202 GPa
Reference modulus of elasticity	203 GPa
Fatigue Stress Limit	48.3 MPa
Material Density	7750 kg/m³
Sensor Average Density	2700 kg/m³
Cyclic Stresses at inline resonance	
Cyclic drag stress $S_{O,max}$	200.7 MPa
Cyclic stress limit	48.2 MPa
Steady state stresses at design velocity	
Calculated Von Mises Stress	0.5 MPa
Max. Allowable Von Mises Stress	206.9 MPa
Dynamic stresses at design velocity	
Combined Drag and Lift Stress	0.4 MPa
Dynamic stress limit	48.2 MPa
Pressure	
Max Allowable Stem Pressure	47.2 MPa
Max Allowable Tip Pressure	414.4 MPa

Assessment Result	PASS

(SBC) Small bore connections (EI AVIFF Guidelines)

Record Id	P&ID	Line reference	Description	SBC Ref.	SBC tag	Fitting type	Length	Valves	SBC diameter	SBC location	Mainline LOF	Mainline schedule	LOF Geom/loc/mod	LOF	Walkdown comments	Survey Link
4	Figure D-1	Stream 7	Downstream of K402	Example D5	Pressure tapping	weldolet	< 600mm	1 (Less than #900)	1'' DN25	mid span	1.00	80	0.66/1.00/0.66	0.66		